Algo HW-2

Q.1. $\log g(n) = \Omega(1)$, $\therefore k > 0$ and n_1

$\therefore \log g(n) \geq k$ for $n \geq n_1$

For $f(n) = O(g(n))$, there exists $c > 0$ and n_0 such that $f(n) \leq c\,g(n)$ for all $n \geq n_0$.

$\log f(n) \leq \log(c(g(n))) = \log c + \log g(n) \Rightarrow n \geq n_0$

$\log c \leq \frac{\log c}{k} \cdot \log g(n)$

where, $\log g(n) \geq k > 0$

$\therefore \log f(n) \leq \log g(n) + \frac{\log c}{k} \log g(n)$

$= \left(1 + \frac{\log c}{k}\right) \log g(n)$

This shows $\log f(n) = O(\log g(n))$

Q. 2. In while $y>0$ loop,
loop is executed n times.

In while $x>0$ loop,
loop is executed $n/2$ times.

$\therefore t = n \cdot [n/2]$

If n is even : $t = n^2/2$
If n is odd : $t = (n^2+n)/2$

$$\therefore \frac{n^2}{2} \leq t \leq \frac{n^2+n}{2} \leq n^2$$

$\therefore O(n^2)$

Q.3. In while $y>0$,
loop is executed $[n/i]$ times

$\therefore t = \sum_{i=1}^{n} [n/i]$

$\therefore \Theta [n/i]$

In for $i=1$ to n,
loop is executed n times

$\therefore t = \sum_{i=1}^{n} [n/i]$

$\therefore t = n \sum_{i=1}^{n} [1/i]$

$\therefore t = n (\log n)$

$\therefore \Theta (n \log n)$

Q.4. ~~For~~ $1 \leq i \leq n$

For while $x \leq i$,

$p \leq i$

$\therefore$ $1+2+4+\cdots+p = 2p-1$

$\therefore i \leq 2p-1 \leq 2i-1$

$$\therefore t = \sum_{i=1}^{n} (2p-1)$$

$$= \frac{n^2+n}{2}$$

$$= \Theta(n^2)$$